13th Batch

Exam Committee

NEU 241 Final OSPE

Disclaimer: A lot of the exam questions were not collected as the exam overall was difficult. **However**, the OSPE section of this exam was the easiest. The questions are very similar to those in other batches and if you just have a look at them there, it should give a general idea about it. All OSPE questions are based 100% on lab material and if you study them you should be able to easily get all the question right. I wish you all the best. <3
– Khaled Hamsho

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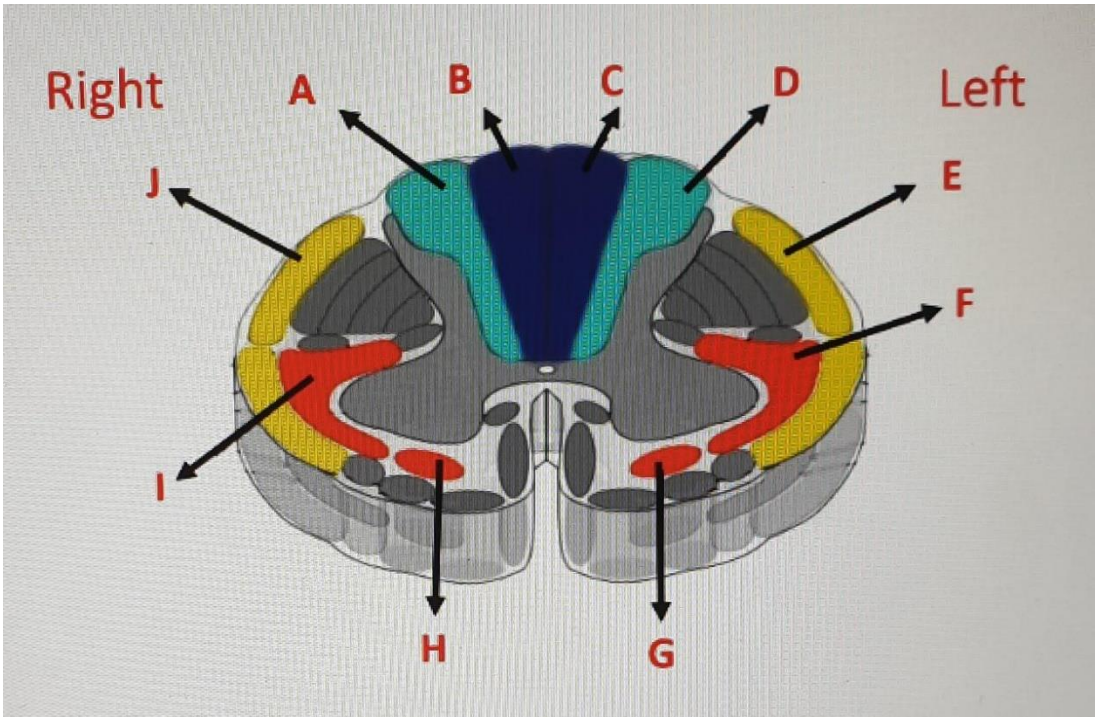
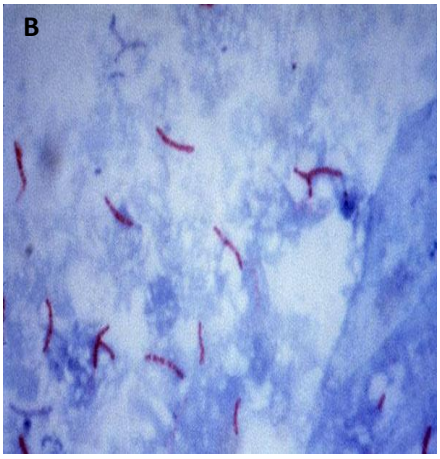
NEU 241 OSPE

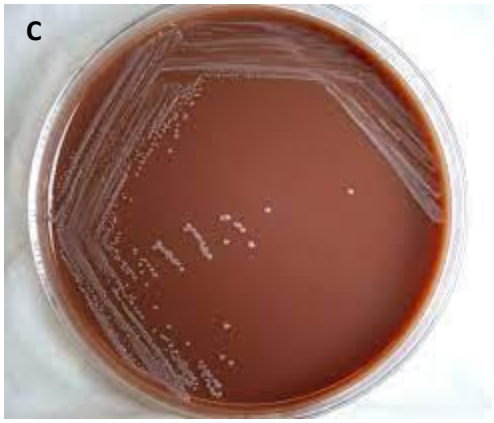
1. On sensory examination of a patient, there is a loss of pain and temperature sensation in his left lower limb. Which of the labelled tracts in the provided spinal cord section transmits pain and temperature?

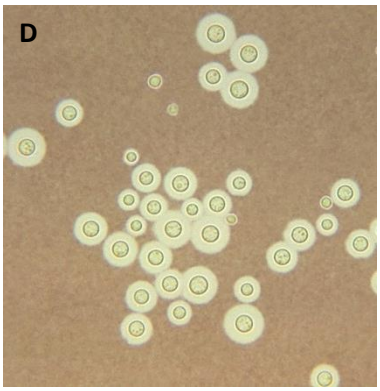


Ans: I

2. Which one of the following shows the satellitism?

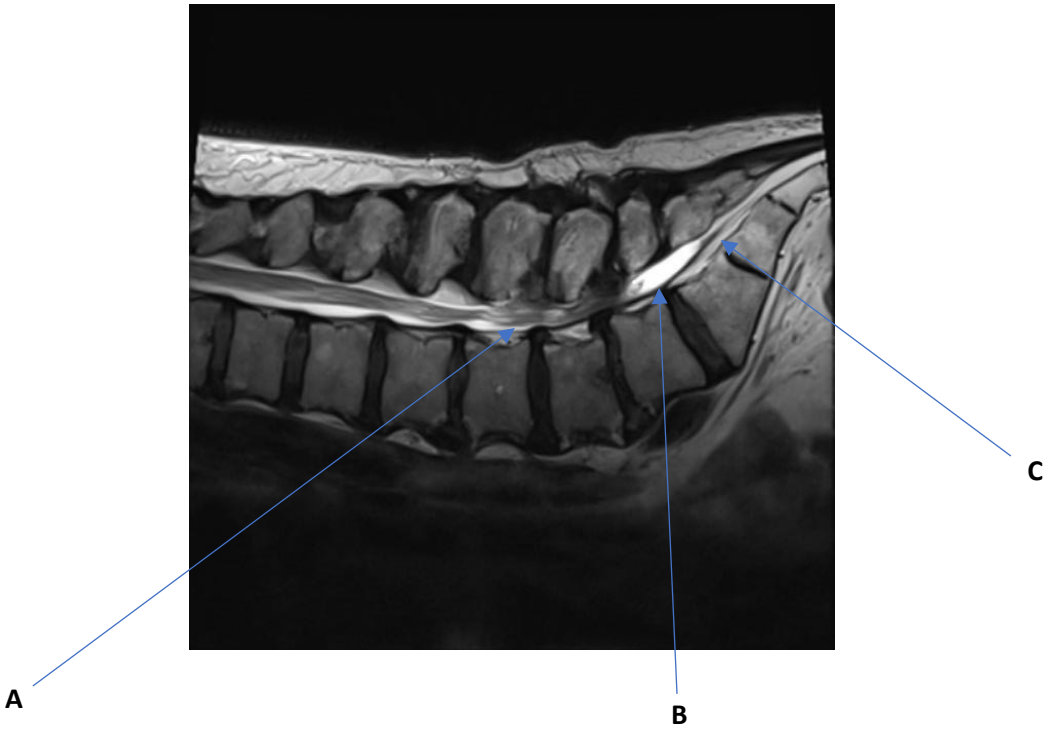






Ans: A

3. At which of the following locations would be ideal to perform a lumbar puncture?



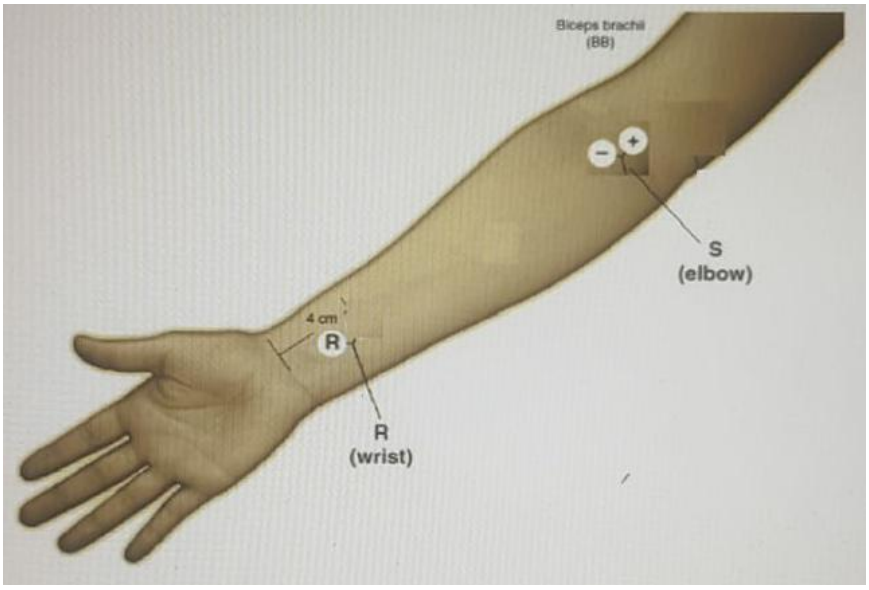
Ans: B

4. A. Identify the circled structure in the image:
   B. State its function

Ans:

A. Node(s) of Ranvier
B. Saltatory conduction of Action Potential

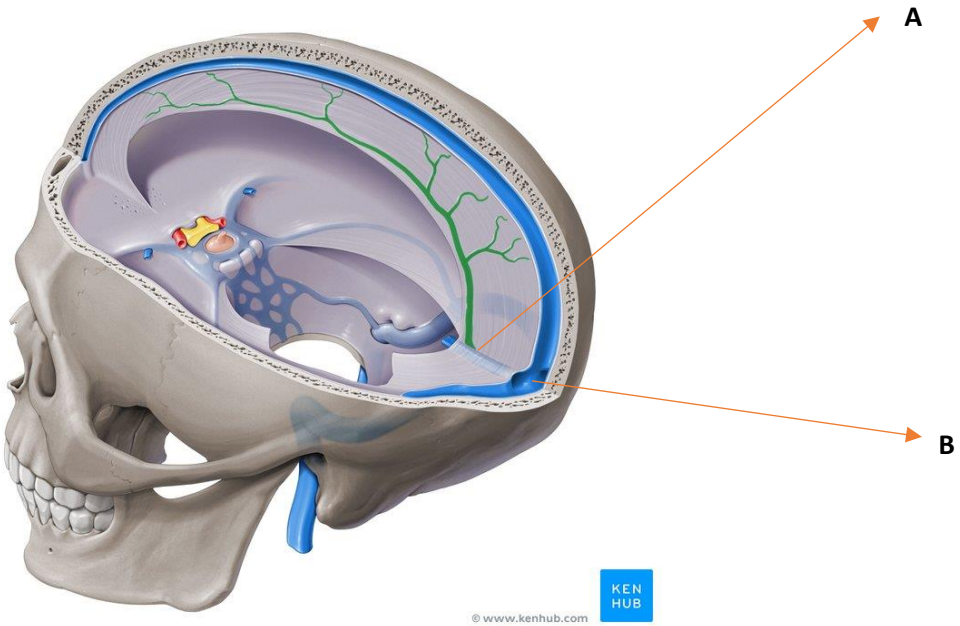
5. The nerve is stimulated at S in the elbow and the response recorded at R in the wrist, 28 cm away. If the latency is 4 milliseconds, calculate the conduction velocity of the nerve in meters per second (m/sec)



A. 7 m/sec
B. 14 m/sec
C. 28 m/sec
D. 70 m/sec
E. 140 m/sec

Ans: D; *(I didn't feel like writing the method to calculate it so I just yeeted it from the 11th Batch)* $\frac{28\ cm \div 100}{4\ millisec \div 1000} = \frac{0.28m}{0.004sec} = 70\ m/sec$

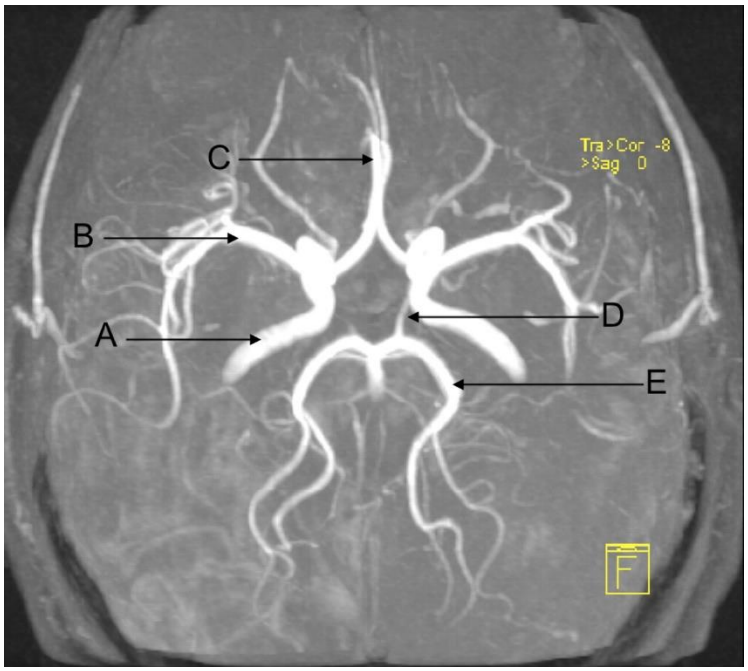
6. The picture shows various dural sinuses *(Not the exact picture in the exam, but I couldn't find anything better)*
   A. Identify structure A
   B. Mention the importance of structure B



Ans:

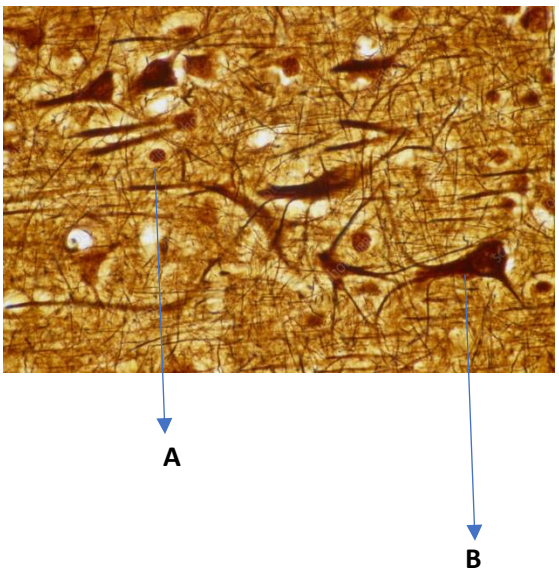
A. Straight sinus
B. Responsible relaying the drainage of most of the brain to the transverse sinuses and eventually the sagittal sinuses into the internal jugular vein (Unsure of answer)

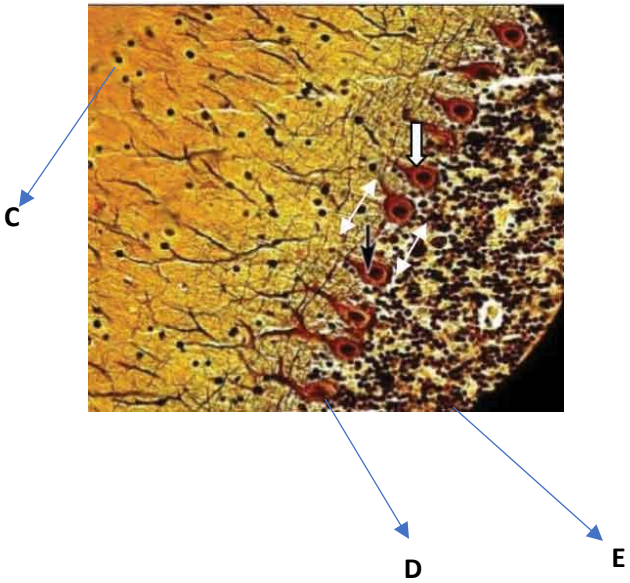
7. Occlusion of which of the following vessels would result in loss of sensation and motor function in the lower limb?



Ans: C

8. Which one of the labeled cells A, B, C, D, or E is responsible for primary motor function of the body?





Ans: B